CHAPTER 1, 3, 4, 5

Scientifice = Fortran - Large number float array

Busines = COBOL - report, use decimal

Artificial = LISP - linked symbol

Systen prog = efficient C

web software = HTMC, PHP

Readability - Writability - Reliability - Cost

Computer architecture - Von Neumann

Data and progran stored memory

memory seperate (CPU) → Control unit
Arithmetic logic unit

Imperative (eml2) = ALL OPP language

functional = LIPS, schene, ML

Logic = Prolog (RULE)

markup = JSTL, XSLT

Reability ↑ cost ↑ writability ↑ readability ↓

writability ↑ readability ↓
C++ pointer

Implementasyon method

- Compilation → large application → slow transla fast exec
- Pure Interpretation → small program → no transla slow execut
- Hybrid System → faster

Compilation = lexical → syntax → semantic → code generat

JUST IN TIME ⇒ Java . Net

Syntax ⇒ form lexical tree (int a)

Semantic = orta kod (int a anlamı)

Lexical = Karakter (sum, int)

Token = tanımlama (semicolon)

John Backus syntax Algo 58

↳ contex free grammer

Describe the syntax → language generation

Ambiguity = 2 farklı anlamı

extended BNF = <expr> → <term> {+|-} <term>

BNF ⇒ <expr> → <expr> + <term>

parse tree ⇒ A hierarchical representation derivation

Compile ⇒ C and Java
Load = static memory
Run time = non static non variable
Static binding = unchanged program execution
Dinamic binding change program execution

SCOPE = Static, dinamic,

Pairwise disjointness test = non left recursive olan

reserved word = dilin içinden ayrılmış kelime int vs...

Left recursive

E: E+T | T
T: T*F | F
F: (E) | id

E: TE'
E': +TE' | ε

T: FT'

T': *FT' | ε

Left derivation bbacaabb

<S> → <A> c <B> b
<A> → b <A> | a
<B> → a <B> | b

<S> → <A> c <B> b
b <A> c <B> b
b b <A> c <B> b
bb a c <B> b
bb a c a a <B> b
bbacaabb

left recursive

$\langle A \rangle \to \langle A \rangle * \langle B \rangle \mid \langle A \rangle / \langle B \rangle \mid a$

$\langle B \rangle \to \langle B \rangle + \langle C \rangle \mid \langle B \rangle - \langle C \rangle \mid b$

$\langle C \rangle \to c$

$\langle A \rangle \to a \langle A' \rangle$

$\langle A' \rangle \to * \langle B \rangle \langle A' \rangle \mid / \langle B \rangle \langle A' \rangle \mid \varepsilon$

$\langle B \rangle \to b \langle B' \rangle$

$\langle B' \rangle \to + + \langle C \rangle \langle B' \rangle \mid - \langle C \rangle \langle B' \rangle \mid \varepsilon$

$\langle C \rangle \to c$

Life times

global = static

local = stack dynamic

pointer = Explic heap dynamic

Static scope

Kod sırasıyla çağrılır

Dinamik scope

Çağrıma sırasına göre